

Tuesday Warm-up: Electric Fields in Matter

Prof. Jordan C. Hanson

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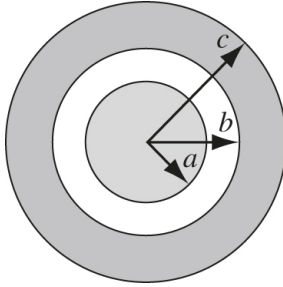


Figure 1: A coaxial cable partially filled with a dielectric material with permittivity ϵ .

1 Memory Bank

- **Relative voltage:**

$$V = - \int_a^b \mathbf{E} \cdot d\mathbf{l} \quad (1)$$

- **The electric displacement:**

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P} \quad (2)$$

- **Gauss' Law for electric displacement:**

$$\oint \mathbf{D} \cdot d\mathbf{a} = Q_{f,enc} \quad (3)$$

- **Linear dielectric materials:**

$$\mathbf{P} = \epsilon_0 \chi_e \mathbf{E} \quad (4)$$

$$\mathbf{D} = \epsilon_0 (1 + \chi_e) \mathbf{E} \quad (5)$$

$$\mathbf{D} = \epsilon \mathbf{E} \quad (6)$$

$$\epsilon_r = \epsilon / \epsilon_0 = 1 + \chi_e \quad (7)$$

- Second, solve for $\mathbf{E}$, knowing that $\mathbf{D} = \epsilon \mathbf{E}$.

- Third, obtain the voltage by integrating $\mathbf{E}$.

- Fourth, use $Q = CV$ to solve for the capacitance, C . What is C in the limit $\epsilon_r \rightarrow 1$?

- Examine Tab. 2. The relationship between the *dielectric constant* and the *index of refraction* is $n \approx \sqrt{\epsilon_r}$. The speed of light is $c \approx 0.3 \text{ m ns}^{-1}$. Suppose a weather satellite radiates a microwave signal through 100 km of space, then 100 km of water vapor, and the signal reflects from the ocean and travels back to the satellite. What is the time duration between transmission and reception? What is the time for signals that travel through air?

2 Linear Dielectric Materials

- Consider Fig. 1. A certain coaxial cable consists of a copper wire, radius a , surrounded by a concentric copper tube of inner radius c . The space between is partially filled (from b out to c) with material of dielectric constant ϵ_r . Find the capacitance per unit length of this cable.

- First, use Gauss' Law, with cylindrical symmetry, to find $\mathbf{D}$ for $a < s < c$.

Material	Dielectric constant	Material	Dielectric constant
Vacuum	1	Benzene	2.28
Helium	1.000065	Diamond	5.7–5.9
Neon	1.00013	Sodium chloride	5.9
Hydrogen (H ₂)	1.000254	Silicon	11.7
Argon	1.000517	Methanol	33.0
Air (dry)	1.000536	Water	80.1
Nitrogen (N ₂)	1.000548	Ice (–30°C)	104
Water vapor (100°C)	1.00589	KTaNbO ₃ (0°C)	34,000

Figure 2: Dielectric properties of different materials.